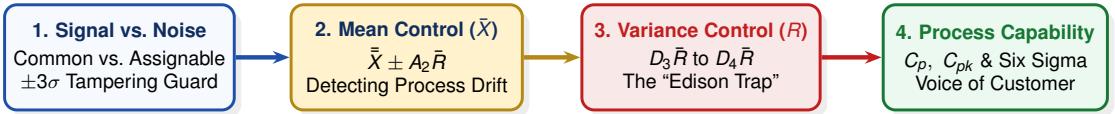
Quality Management, Statistical Process Control (SPC) & Process Capability

The Theranos Quality Forensics Lab, AI Process Monitoring & Six Sigma Capability



"Why did a $9 billion Silicon Valley blood-testing unicorn collapse when CMS inspectors audited just 6 months of quality control logs—and how do statistical control charts separate normal operational noise from catastrophic process failure?"

Session Roadmap: 4 Forensics Rounds, 4 Theory Blocks & 4 Interactive Web Labs

ROUND 1: SIGNAL vs. NOISE

- **Theranos CMS Audit:** Newark Lab QC Collapse
- **Cost of Quality (COQ):** 1–10–100 Rule & Deming
- **Theory 1:** $\pm 3\sigma$ (2,700 DPMO) & $\sigma_{\text{tamp}} = \sqrt{2}\sigma$ Trap
- **Sim Round 1 (Q1–Q6):** Glucose QC Tampering (**11B**)

ROUND 2: MEAN ($\bar{X}$)

- **Subgroup Sampling:** Control vials n vs. batches N
- **CLT & Range Shortcut:** $\sigma_{\bar{X}} = \sigma/\sqrt{n}$, $\hat{\sigma} = \bar{R}/d_2$
- **Theory 2:** $\bar{\bar{X}} \pm A_2\bar{R}$ Immunoassay & Run Rules
- **Sim Round 2 (Q7–Q12):** Potassium Andon Cord (**22B**)

ROUND 3: VARIANCE (R)

- **Why $\bar{X}$ Misses Spread:** Batches 101–106 (**26B**)
- **Theory 3:** R-Chart Limits $[D_3\bar{R},\ D_4\bar{R}]$ & Golden Rule
- **Sim Round 3 (Q13–Q18):** B12 Contaminated Pool (**33B**)
- **Coagulation QC Audit:** $N = 6$, $n = 20$ PT/INR (**37B**)

ROUND 4: CAPABILITY

- **Analyzer vs. CLIA Spec:** [LCL, UCL] vs. [LSL, USL]
- **Theory 4:** C_p vs. C_{pk} & Six Sigma 1.5σ Shift
- **Sim Round 4 (Q19–Q24):** CMS \$100k Portfolio (**45B**)
- **Poll & Quiz Q1–Q24:** Master Key (**48B, 50B**)

Pedagogical Rhythm Across All 4 Clinical Forensics Modules

WSJ / CMS Clinical Forensics $\longrightarrow$ SPC Mathematical Theory $\longrightarrow$ Interactive Web Simulator & Gemini AI (Canvas Quiz Q1–Q24) $\longrightarrow$ Clinical Executive Action.

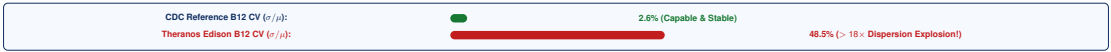
The $9 Billion Quality Collapse: Inside the Theranos Newark Lab

The Newark, CA Lab Audit (Oct 2014 – Jun 2015)

- Valuation hit **$9 billion** on promises to run 200+ lab tests from a tiny finger-stick drop on the proprietary **Edison** analyzer.
- Behind closed doors, tests were split between unstable **Edison** devices and commercial **Siemens** machines.
- CMS federal inspectors audited 6 months of Statistical Process Control (SPC) logs and uncovered systematic quality collapse.

Four Smoking-Gun QC Failures (WSJ / CMS Report)

1. **Prolactin hormone assay:** Failed daily quality-control checks **47% of the time** over 4 months.
2. **PSA (Prostate Cancer) assay:** Failed **22% of QC checks** on Edison over 6 months.
3. **Vitamin B12 dispersion:** Coefficient of Variation ($\text{CV} = \sigma/\mu$) hit **34.3%–48.5%** vs. CDC benchmark of **2.4%–2.9%** ($> 15\times$ noisier!).
4. **Moving the Goalposts:** Management deleted outlier logs and rewrote "acceptable criteria to match the data."



Dimensions of Quality & The Cost of Quality (COQ) Iceberg in Diagnostics

Design Quality vs. Conformance Quality

- **Design Quality (Assay Architecture):**
  - Target clinical features & diagnostic promise.
  - *Theranos Promise:* Finger-stick 25 μL nanotainer micro-volume blood panel at Walgreens clinics.
  - *Reference Lab:* Standard 5 mL venous blood draw on automated hospital analyzers.
- **Conformance Quality (SPC's Focus):**
  - How consistently actual assay output meets CLIA regulatory accuracy limits (μ and σ).
  - *Potassium:* Within ± 0.5 mmol/L of target.
  - *Calcium:* Within $[8.50,\ 10.50]$ mg/dL tolerance.

1. Prevention Costs ($)
- Validated optical/pipette R&D
- CLIA technician SPC SOPs
- Automated reagent refrigeration

2. Appraisal Costs ($$)
- Shift QC subgroup sampling (n)
- Running High/Low plasma vials
- Daily $\bar{X}$ and R-chart audits

3. Internal Failure ($$$)
- Re-running failed Prolactin QC
- Scrapped Edison cassettes
- Halted Newark assay shifts

4. External Failure ($$$$$)
- ER bleed/potassium misdiagnoses
- 810,000 voided patient tests
- CLIA revocation & $9B wipeout

The 1–10–100 Rule of Clinical Quality Economics: **$1** invested in Assay Prevention saves **$10** in QC Re-Runs and **$100+** in Patient Harm & Regulatory Collapse.

Clinical Assay Variation: Why Every Blood Test Exhibits Natural Noise

1. Immunoassay Photonics



Prolactin & PSA Vials

Photomultiplier tube photon counts vary slightly across identical preserved plasma vials.

2. Micro-Fluidic Pipetting



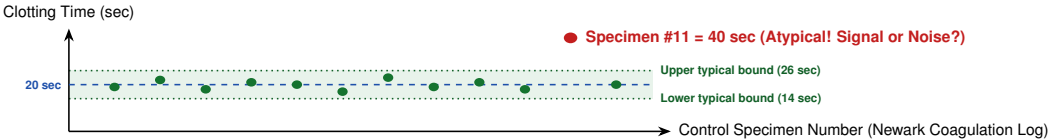
Edison Nanotainer Drops

Finger-stick dilution of 25 μL whole blood introduces micro-volume dispersion vs. venous draws.

3. Coagulation Clotting (PT/INR)



Prothrombin clotting assays average **20.0 sec** on normal plasma—what if Specimen #11 takes **40.0 sec**?



Common-Cause Variation (Noise) vs. Assignable-Cause Variation (Signal)

1. Common-Cause / Natural Assay Noise

- **Clinical Source:** Inherent to analyzer physics (normal optical sensor shot noise, $\pm 0.2^{\circ}$C incubator micro-fluctuations, natural reagent luminescence decay within shelf life).
- **Behavior:** Forms a stable distribution $N(\mu_0, \sigma_0^2)$. **Assay is In Statistical Control.**
- **Who Owns It? R&D / Biomedical Engineering** (requires better optics/fluidics to shrink σ_0; lab techs *cannot* fix it by twisting calibration offsets).

2. Assignable / Special-Cause Assay Signal

- **Clinical Source:** Specific operational failure (unrefrigerated/expired reagent lot, air bubble or hemolyzed clot in finger-stick nanotainer, uncalibrated Edison robotic pipette arm).
- **Behavior:** Shifts assay mean ($\mu_0 \to \mu_1$) or inflates spread ($\sigma_0 \to \sigma_1$). **Assay is Out of Control.**
- **Who Owns It? Frontline Lab Technicians / CLIA QA** (halt patient testing immediately, quarantine batch, fix root cause).



Deming's Funnel & The "Tampering Trap": Why Chasing Noise Doubles Variance

The Over-Calibrating Technician Mistake

- A blood analyzer is **in control** at target $\mu = 100$ mg/dL with pure random noise $\varepsilon_t \sim N(0, \sigma^2)$.
- At run $t-1$, natural error is $\varepsilon_{t-1} = +4$ ($Y_{t-1} = 104$).
- **Tampering Reflex:** A nervous technician dials the calibration offset down by $-\varepsilon_{t-1} = -4$ for run t.
- Run t now experiences both the -4 adjustment *and* fresh noise ε_t:

$$Y_t^{\text{tamp}} = \mu + \varepsilon_t - \varepsilon_{t-1}$$

The Mathematical Penalty of Tampering

Since ε_t and ε_{t-1} are independent with variance σ^2:

$$\text{Var}(Y_t^{\text{tamp}}) = \text{Var}(\varepsilon_t - \varepsilon_{t-1}) = \sigma^2 + \sigma^2 = \mathbf{2}\sigma^2$$

$$\implies \sigma_{\text{tampered}} = \sqrt{\mathbf{2}}\,\sigma \approx \mathbf{1.414}\sigma!$$

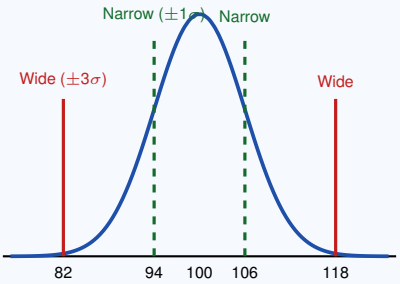
- Chasing random noise inside an in-control assay **doubles process variance (+100%)** and inflates standard deviation by +41.4%!

Deming Funnel Rule 1 (Hands Off Inside $\pm 3\sigma$):
Leave aimed at target $\mu \implies \text{Var}(Y_t) = \mathbf{1.0}\sigma^2$ (SD = 1.
Minimum possible variance achievable by current analyzer h

Deming Funnel Rule 2 (Compensating Every QC Run):
Adjust by $-\varepsilon_{t-1}$ every run $\implies \text{Var}(Y_t) = \mathbf{2.0}\sigma^2$ (SD = 1.414σ)
Well-intentioned technician recalibration manufactures chaos!

Setting Control Limits: Type I (α, False Alarm) vs. Type II (β, Missed Shift)

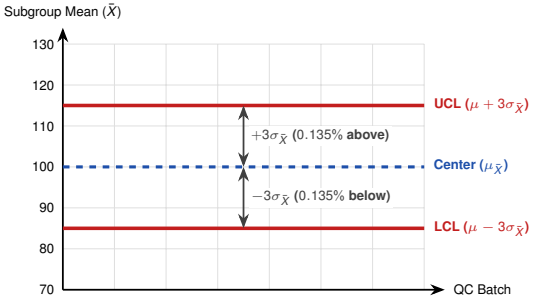
Narrow vs. Wide Control Limits



The Clinical SPC Trade-Off Matrix

- **Type I Error (α — False Alarm / "Crying Wolf"):**
  - Concluding the blood analyzer is out of control when **only common-cause shot noise** is present.
  - *If limits are too narrow ($\pm 1\sigma$):* $\alpha = 31.7\%$! Techs halt the lab every 3rd QC vial and tamper with calibration.
- **Type II Error (β — Missed Shift / "Sleeping Watchdog"):**
  - Concluding the analyzer is in control when a **real reagent or pipette failure** has occurred.
  - *If limits are too wide ($\pm 5\sigma$):* Zero false alarms, but misdiagnosed patient blood panels ship to physicians for days undetected.

Why $\pm 3\sigma$ Is the Global SPC Standard (99.73% Coverage & 2,700 DPMO)



Mathematical & Spreadsheet Representation ($\pm 3\sigma$)

1. **Cumulative Normal Probabilities ($Z = \pm 3$):**
   - $\Phi(+3) =$ `=NORM.S.DIST(3, TRUE)` $=$ **0.99865**
   - $\Phi(-3) =$ `=NORM.S.DIST(-3, TRUE)` $=$ **0.00135**
2. **False-Alarm Probability Outside $\pm 3\sigma$ (α):**

   $$P(|Z| > 3) = 0.00135 + 0.00135 = \mathbf{0.0027}\ (\mathbf{0.27}\%)$$

3. **False Alarms / Defects Per Million (DPMO):**

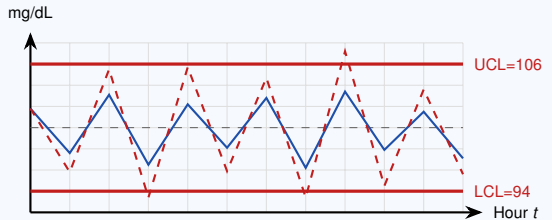
   $10^6 \times 0.0027 =$ **2,700** per million ($\approx$ 1 in 370 QC runs)

Visual Intuition (Before Scenario 1): Analyzer Deadbands vs. Tampering

Siemens Deadband vs. Newark Knob-Twisting

- A calibrated fasting-glucose assay ($\mu = 100.0$ mg/dL, $\sigma = 2.0$ mg/dL) exhibits normal shot noise across hourly QC vials.
- **Policy 1 — CLIA-Compliant Commercial Analyzer (Siemens):** Locks calibration inside the $\pm 3\sigma$ deadband [94.0, 106.0] mg/dL; ignores random common-cause noise.
- **Policy 2 — Newark Lab Manual Recalibration ("Tampering"):** Technicians manually shift the calibration offset by $-\varepsilon_{t-1}$ after every single QC vial reading!

Preview: Untampered (σ) vs. Tampered ($\sqrt{2}\sigma$)



Solid Blue: Siemens Deadband stays inside [94, 106]. **Dashed Red:** Newark Tampering breaches limits!

LAB SESSION 1: Technician Tampering vs. $\pm 3\sigma$ Deadband (Quiz Q1–Q6)

Open Web Simulator Round 1 & Answer Canvas Quiz Q1–Q6

Newark 12-hour Glucose QC log
(target $= 100.0,\ \sigma = 2.0,$ [94.0, 106.0] mg/dL;
\$500/recal, \$5,000/breach):

- **Q1 (Policy A — $\pm 3\sigma$ Deadband):** Sample SD s (2.83) and 12-hr spread R (7.20 mg/dL)?
- **Q2 (Policy A Stability):** True/False: 0 breaches, 0 recalibrations, and \$0 total quality cost?
- **Q3 (Policy B — Full Tampering $Y_t = 100 + \varepsilon_t - \varepsilon_{t-1}$):** What do s and range R jump to?
- **Q4 (Policy B Cost):** How many hours breach [94.0, 106.0] and what is total quality cost?
- **Q5 (Policy C — Narrow $\pm 1\sigma$ Deadband** [98.0, 102.0]**):** Recalibrations, breaches, and cost?
- **Q6 (Deming Synthesis):** Why do Policies B & C inflate variance ($\sigma_{\text{tamp}} = \sqrt{2}\sigma$) and Type I α?

Web Simulator + 1-Click Gemini AI Prompt (Round 1)

```
Open theranos-sim.html (embedded in Canvas or GitHub Pages) ->
Round 1.
Toggle Policies A (+/-3 sigma), B (Full Tampering), and C (+/-1
sigma) or click "Copy Scenario Prompt for Gemini":
Natural errors e = [1.8, -2.4, 3.1, -3.5, 2.2, -1.9, 2.8, -3.8,
3.4, -2.1, 1.5, -2.9].

Compute mean, sample SD (ddof=1), range, recalibration count ($500

each), breach count outside [94, 106] ($5,000 each), and Total

Quality Cost for Policies A, B, and C to answer Q1--Q6.
```

Under the Hood (Simulator / Sheets / Python)

- **Sheets:** `Y_t = 100 + B3 - B2` | `=STDEV.S(C2:C13)`
- **Python:** `y1 = 100+e; y2 = 100+e-np.r_[0,e[:-1]]; np.std(y2,ddof=1)`

FORMULA REMINDER: 3σ Deadband $= \mu \pm 3\sigma =$ [94.0, 106.0] | Tampered Output: $Y_t^{\text{tamp}} = \mu + \varepsilon_t - \varepsilon_{t-1} \implies \sigma_{\text{tamp}} = \sqrt{2}\sigma \approx 1.414\sigma$

Module 1 Executive Synthesis: The Two Laws of Statistical Process Control

Law 1: Don't Tamper With Common-Cause Noise

- When sample points fall **inside** [LCL, UCL], the process is exhibiting natural variation (μ_0, σ_0).
- **Operational Rule:** Hands off the calibration knobs! Reacting to random noise inflates σ by $\sqrt{2}$ ($+41.4\%$).
- **How to Improve:** Only capital/engineering redesign (better optics, automated temperature control) can shrink common-cause σ_0.

Law 2: Act Immediately on Assignable Signals

- When a sample point breaches **outside** [LCL, UCL], there is only a 0.27% (2,700 per million) chance it happened by luck.
- **Operational Rule:** Stop the line immediately, quarantine recent output, and fix the special cause.
- **Never "Move the Goalposts":** Deleting outliers or widening limits turns quality control into fraud.

Bridge to Module 2: Why don't we plot individual items X_i on a control chart? Why do we group items into periodic **subgroups of size** n and plot the **Sample Mean** ($\bar{X}$) and **Sample Range** (R)?

Controlling the Process Mean ($\bar{X}$-Charts)

Subgroup Sampling (n vs. N), Central Limit Theorem & The A_2 Formula

Clinical Quality Control by Subgroup Sampling (n vs. N)

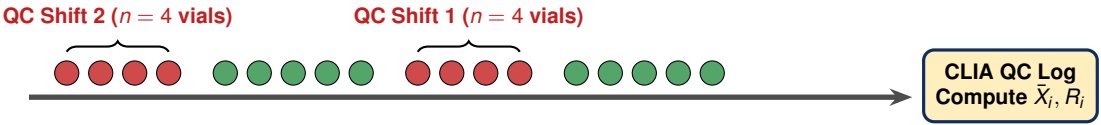
Dual Mission of Clinical Subgroup SPC

Periodically run a subgroup of preserved blood-plasma control vials to detect:

1. A **shift in the assay mean** ($\bar{X}$-chart, e.g., reagent drift), or
2. An **increase in assay dispersion** (R-chart, e.g., pipette dilution noise).

CLIA Subgroup Protocol (n vs. N)

- **Sample size (n):** Number of preserved control vials measured *within each QC shift/batch* (e.g., $n = 4$ glucose plasma vials per shift).
- **Number of samples (N):** Total number of QC shifts/batches audited over time (e.g., $N = 6$ consecutive shifts).



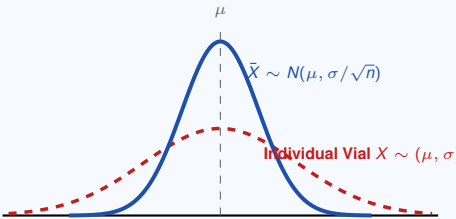
Why Sample Means ($\bar{X}$)? The Central Limit Theorem Advantage

Two Statistical Superpowers of Averaging n Vials

1. **Normality Guarantee (CLT):** Even if individual assay readings X are skewed (e.g., trace hormone biomarkers or clotting times), the subgroup mean $\bar{X}$ is approximately **Normally distributed**—so $\pm 3\sigma_{\bar{X}}$ always covers 99.73%!
2. **Standard Error Compression ($\sigma/\sqrt{n}$):** Averaging n control vials cancels random noise, tightening the standard deviation of $\bar{X}$:

$$\mu_{\bar{X}} = \bar{\bar{X}} = \frac{1}{N}\sum_{i=1}^{N} \bar{X}_i, \qquad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Individual Vials (X) vs. Subgroup Means ($\bar{X}$)



For $n = 4$ vials, $\sigma_{\bar{X}} = \sigma/\sqrt{4} = 0.5\sigma$ (twice as sharp at catching reagent drift!).

Anatomy of Clinical Subgroup Data: Glucose QC Vials ($\bar{X}_i$ & R_i)

Shift (i)	Preserved Glucose Plasma Vials ($n = 4$ vials/shift, in mg/dL)				Mean ($\bar{X}_i$)	Range ($R_i = \max - \min$)
1	97.06	97.32	100.19	101.02	**98.90**	$101.02 - 97.06 = \mathbf{3.96}$
2	97.68	100.05	97.80	97.85	**98.34**	$100.05 - 97.68 = \mathbf{2.37}$
3	98.76	98.22	100.72	103.11	**100.20**	$103.11 - 98.22 = \mathbf{4.89}$
4	101.34	100.00	103.19	105.79	**102.58**	$105.79 - 100.00 = \mathbf{5.79}$
5	102.78	105.28	105.02	97.96	**102.76**	$105.28 - 97.96 = \mathbf{7.32}$
6	104.23	104.57	97.50	100.39	**101.67**	$104.57 - 97.50 = \mathbf{7.07}$

60-Second Quick Check

Suppose **Sample 7** ($n = 4$ preserved glucose plasma vials, in mg/dL) has readings `[99.10, 102.50, 98.40, 100.00]`. What are $\bar{X}_7$ and R_7, and what are the baseline averages $\bar{\bar{X}}$ and $\bar{R}$ across Shifts 1–6 ($N = 6$)?

FORMULA REMINDER: $\bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} X_{ij}$ | $R_i = \max_j(X_{ij}) - \min_j(X_{ij})$ | $\bar{\bar{X}} = \frac{1}{N}\sum_{i=1}^{N} \bar{X}_i$ | $\bar{R} = \frac{1}{N}\sum_{i=1}^{N} R_i$

Estimating σ from Sample Ranges ($\bar{R}/d_2$) & Deriving the A_2 Shortcut

Step 1: Estimating Assay σ via $\bar{R}$

- In clinical SPC, the true within-batch assay standard deviation σ is estimated from the **average of the subgroup ranges** ($\bar{R}$):

$$\hat{\sigma} = \frac{\bar{R}}{d_2} \qquad \text{where} \qquad \bar{R} = \frac{1}{N}\sum_{i=1}^{N} R_i$$

- d_2 is Hartley's statistical constant (depends *only* on subgroup size n). For small QC subgroups ($n \leq 10$ vials), range R_i is fast and nearly as efficient as sample SD s_i.

Step 2: Bundling $3/(\sqrt{n}\,d_2)$ into A_2

Substitute $\sigma_{\bar{X}} = \dfrac{\sigma}{\sqrt{n}} = \dfrac{\bar{R}}{\sqrt{n}\,d_2}$ into $\bar{\bar{X}} \pm 3\sigma_{\bar{X}}$:

$$\text{UCL}_{\bar{X}} = \bar{\bar{X}} + \underbrace{\left(\frac{3}{\sqrt{n}\,d_2}\right)}_{A_2}\bar{R} = \bar{\bar{\mathbf{X}}} + \mathbf{A_2}\bar{\mathbf{R}}$$

$$\text{LCL}_{\bar{X}} = \bar{\bar{X}} - \underbrace{\left(\frac{3}{\sqrt{n}\,d_2}\right)}_{A_2}\bar{R} = \bar{\bar{\mathbf{X}}} - \mathbf{A_2}\bar{\mathbf{R}}$$

Master $\bar{X}$-Chart Control Limit Equation

$$\text{Control Limits}_{\bar{\mathbf{X}}} = \bar{\bar{\mathbf{X}}} \pm \mathbf{A_2}\bar{\mathbf{R}} \qquad \left(\text{Center Line} = \bar{\bar{X}},\ \text{Half-Width} = A_2\bar{R} = 3\sigma_{\bar{X}}\right)$$

Table of A_2 Constants & The Classic "n vs. N" Exam Trap

The #1 Exam & Case Trap: n vs. N

Sample size (n) is the number of control vials **in each subgroup**—and **NEVER** the number of batches (N)!

- If you audit $N = 6$ batches and each batch has $n = 5$ control vials, look up $\mathbf{n = 5} \implies \mathbf{A_2 = 0.58}$ (do *not* look up 6!).

Why Does A_2 Decrease as n Increases?

Because $A_2 = \dfrac{3}{\sqrt{n}\, d_2}$: larger n shrinks $\sigma_{\bar{X}} = \sigma/\sqrt{n}$ **and** increases d_2, making batch averages much tighter around $\bar{\bar{X}}$!

Subgroup Size (n)	Constant d_2	Factor A_2
2	1.128	1.88
3	1.693	1.02
4	2.059	0.73
5	**2.326**	**0.58**
6	2.534	0.48
7	2.704	0.42
8	2.847	0.37
9	2.970	0.34
10	3.078	0.31
15	3.472	0.22
20	3.735	0.18

Walkthrough: Immunoassay QC Baseline Limits ($N = 6$ Batches, $n = 5$ Vials)

Batch	Validated Analyzer Ferritin / Hormone QC Vials ($n = 5$, in ng/mL)					Mean ($\bar{X}$)	Range (R)
1	49.59	53.94	47.11	60.58	43.48	50.94	17.10
2	60.41	48.90	29.19	58.87	39.80	47.44	31.22
3	65.89	56.96	44.15	60.76	50.39	55.63	21.74
4	48.48	51.02	57.36	45.54	63.11	53.10	17.56
5	49.10	40.69	42.10	29.26	46.40	41.51	19.84
6	42.76	35.11	47.66	46.78	44.93	**43.45**	**12.55**
Baseline Grand Averages across $N = 6$ batches (ng/mL):						$\bar{\bar{X}} = 48.68$	$\bar{R} = 20.00$

Step-by-Step $\bar{X}$-Chart Control Limits ($n = 5$ vials $\implies A_2 = 0.58$)

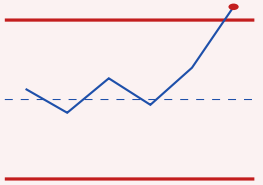
- **Center Line ($\bar{\bar{X}}$):** $\bar{\bar{X}} = \dfrac{50.94 + 47.44 + 55.63 + 53.10 + 41.51 + 43.45}{6} = \mathbf{48.68}$ ng/mL
- **Upper Control Limit ($UCL_{\bar{X}}$):** $\bar{\bar{X}} + A_2\bar{R} = 48.68 + (0.58)(20.00) = 48.68 + 11.60 = \mathbf{60.28}$ ng/mL
- **Lower Control Limit ($LCL_{\bar{X}}$):** $\bar{\bar{X}} - A_2\bar{R} = 48.68 - (0.58)(20.00) = 48.68 - 11.60 = \mathbf{37.08}$ ng/mL

Reading Control Charts Like a Pro: Western Electric / Nelson Run Rules

Why Wait for a $\pm 3\sigma$ Breach When Subtle Assay Drift Is Already Visible?

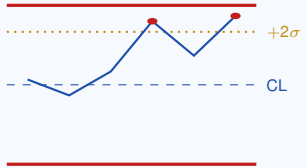
Even before a single batch crosses UCL or LCL, non-random patterns inside [LCL, UCL] reveal an assignable clinical shift in progress:

Rule 1: 1 Point Beyond $\pm 3\sigma$



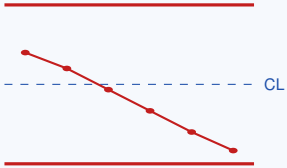
Immediate Out-of-Control Breach ($p = 0.0027$). Halt patient testing!

Rule 2: 2 of 3 Beyond $\pm 2\sigma$



Zone A Early Warning: 2 of 3 consecutive batches past 2σ on same side.

Rules 3 & 4: 8 One-Sided / 6 Trend



Sustained Shift / Drift: 8 batches on one side of CL ($0.5^8 = 0.0039$) or 6 trending.

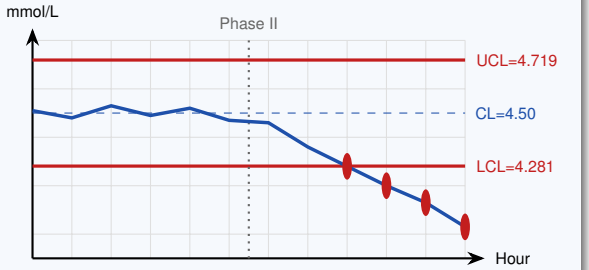
Clinical Forensics: In CMS audits, Theranos logs showed steady downward trends from reagent decay hours before crossing LCL.

Visual Intuition (Before Scenario 2): Potassium Reagent Drift on $\bar{X}$

Clinical Mechanism: Enzyme / Reagent Decay

- In automated blood tests (Potassium / Prolactin), chemical reagents degrade when left unrefrigerated.
- **Phase I Baseline (Hours 1–6):** Calibrated analyzer ($n = 4$ vials/hr) establishes $\bar{\bar{X}} = 4.50$ mmol/L and $\bar{R} = 0.30$ mmol/L.
- **Phase II Monitoring (Hours 7–12):** At Hour 7, a degraded reagent lot is loaded; subgroup means drift steadily downward hour after hour.

Phase I Baseline vs. Phase II Downward Drift



Both **Rule 1** (LCL breach at Hour 9) and **Rule 4** (6-point downward trend) flag the bad reagent!

LAB SESSION 2: Potassium Reagent Drift & Andon Cord (Quiz Q7–Q12)

Open Web Simulator Round 2 & Answer Canvas Quiz Q7–Q12

Newark Potassium Assay ($n = 4$ vials/hr $\implies$ $A_2 = 0.73$; Phase I baseline $\bar{\bar{X}} = \mathbf{4.50}$, $\bar{R} = \mathbf{0.30}$ mmol/L):

- **Q7 (Phase I Limits):** Compute CL, $\text{UCL}_{\bar{X}}$ (4.719), and $\text{LCL}_{\bar{X}}$ (4.281 mmol/L).
- **Q8 (n vs. N Trap):** True/False: Using $N = 6$ ($A_2 = 0.48$) yields overly narrow limits [4.356, 4.644]?
- **Q9 (First Breach Hour):** At which exact Hour (7–12) does $\bar{X}_t$ first fall below $\text{LCL}_{\bar{X}} = 4.281$?
- **Q10 (Cost of Never Stopping):** Out-of-control hours released and total \$20k/hr CMS penalty?
- **Q11 (Optimal Andon Cord Pull):** Which stopping hour minimizes Total Cost of Quality (\$3,000)?
- **Q12 (Run Rules & CLT):** $\hat{\sigma} = \bar{R}/d_2 = 0.1457$, $\sigma_{\bar{X}} = 0.0729$, and 6-point downward trend rule.

Web Simulator + 1-Click Gemini AI Prompt (Round 2)

```
Open theranos-sim.html -> Round 2 (Potassium Andon Cord).
Test pulling the Andon Cord at Hours 7..12 or Never, or click
"Copy Scenario Prompt for Gemini":
Phase I (Hours 1-6, n=4 -> A2=0.73, d2=2.059):  X_double_bar =
4.500, R_bar = 0.300.
Phase II (Hours 7-12) X_bar = [4.460, 4.360, 4.280, 4.200, 4.130,
4.030], R = [0.17, 0.15, 0.19, 0.18, 0.17, 0.21].

Costs:  $3,000 cassette + $5,000 if pulled before LCL breach +

$20,000 per OOC hour released.  Answer Q7--Q12.
```

Under the Hood (Simulator / Sheets / Python)

- **Sheets:** `UCL = 4.50 + 0.73*0.30` | `LCL = 4.50 - 0.73*0.30`
- **Python:** `means = np.mean(data, axis=1); lcl = 4.50 - 0.73*0.30`

FORMULA REMINDER: $n = 4 \implies A_2 = 0.73$ | $\text{UCL}_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}$ | $\text{LCL}_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R}$ | Breach if $\bar{X}_i \notin [\text{LCL}_{\bar{X}}, \text{UCL}_{\bar{X}}]$

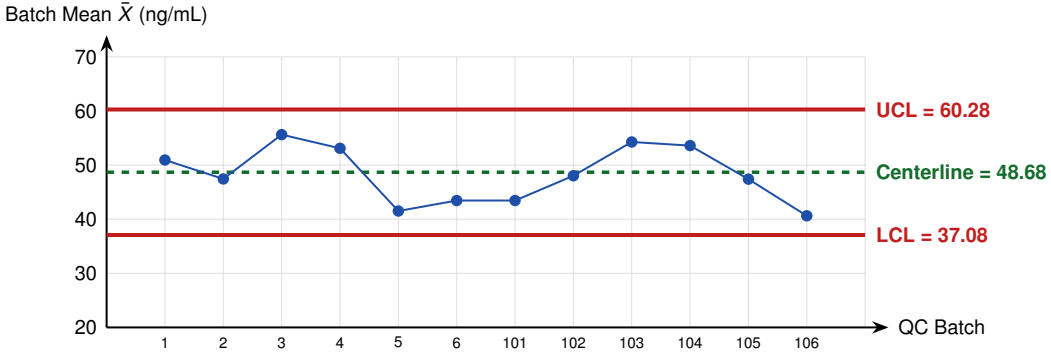
Edison Transfer Mystery: Immunoassay Batches 101–106 (In Control?)

Batch	Edison Micro-Cassette Ferritin / Hormone QC Vials ($n = 5$, $\text{LCL}_{\bar{X}} = 37.08$, $\text{UCL}_{\bar{X}} = 60.28$ ng/mL)					Mean ($\bar{X}$)
101	36.62	50.01	44.39	35.84	50.41	**43.45**
102	42.91	34.72	54.82	51.67	56.01	**48.03**
103	57.50	49.69	69.49	43.18	51.50	**54.27**
104	58.36	53.28	55.59	59.06	41.70	**53.60**
105	30.22	**16.00**	51.87	44.67	**94.27**	**47.40**
106	**2.92**	28.45	**76.67**	66.05	29.04	**40.63**

$\bar{X}$-Chart Evaluation ($\text{LCL}_{\bar{X}} = 37.08,\ \text{UCL}_{\bar{X}} = 60.28$ ng/mL)

Transferred to Edison micro-cassettes, all six batch means ($\bar{X}_{101} \ldots \bar{X}_{106}$) sit inside [37.08, 60.28] ng/mL. *Yet Batches 105 (16.00 to 94.27 ng/mL) and 106 (2.92 to 76.67 ng/mL) suffer wild micro-pipette dilution erraticism!*

Plotting the Batch Means ($\bar{X}$-Chart: Batches 1–6 vs. Edison 101–106)

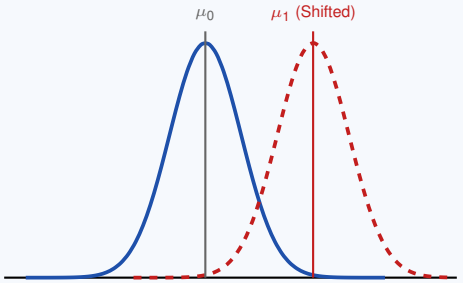


The Blind Spot of the $\bar{X}$-Chart on Edison Micro-Cassettes

On the $\bar{X}$-chart alone, Edison Batches 105 ($\bar{X} = 47.40$ ng/mL) and 106 ($\bar{X} = 40.63$ ng/mL) hide right inside $[37.08,\ 60.28]$—Batch 105 sits right next to the centerline 48.68 ng/mL!

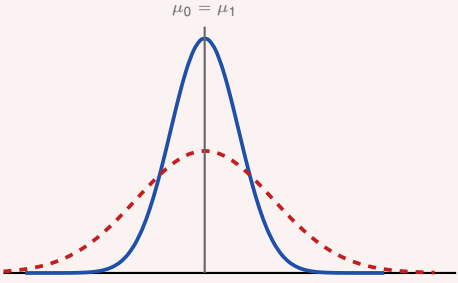
Why $\bar{X}$-Charts Capture Mean Shifts But Miss Variance Explosions

$\bar{X}$-Chart Captures Mean Shift ($\mu_0 \to \mu_1$)



Distribution shifts sideways; subgroup averages $\bar{X}_i$ rapidly cross $\text{UCL}_{\bar{X}}$ or $\text{LCL}_{\bar{X}}$.

$\bar{X}$-Chart Misses Dispersion Spike ($\sigma_0 \to 3\sigma_0$)!



Extreme highs (`94.27`) and extreme lows (`16.00`) **cancel each other out** inside $\bar{X}_i$, hiding a catastrophic variance explosion!

Measuring Shifts in Variability: Range Charts (R-Charts)

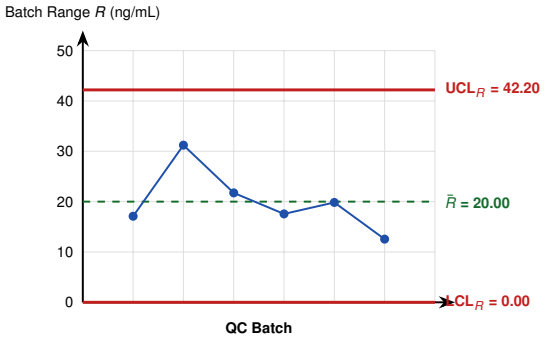
Quick Check on Edison Immunoassay Batches 105 & 106

Look back at **Batch 105** (`[30.22, 16.00, 51.87, 44.67, 94.27]` ng/mL) and **Batch 106** (`[2.92, 28.45, 76.67, 66.05, 29.04]` ng/mL):

1. Why did $\bar{X}_{105} = 47.40$ ng/mL completely mask the micro-pipette dilution failure on the $\bar{X}$-chart?
2. Compute the Batch Ranges R_{105} and R_{106} and compare them against our validated analyzer baseline range $\bar{R} = 20.00$ ng/mL.

FORMULA REMINDER: $\bar{X}_i = \frac{1}{n}\sum_{j=1}^{n} X_{ij}$ (cancels symmetric $+$ and $-$ outliers) | $R_i = \max_j(X_{ij}) - \min_j(X_{ij})$ (captures total spread)

Range Chart (R-Chart) Architecture: $\bar{R}$, $\text{UCL}_R = D_4\bar{R}$, and $\text{LCL}_R = D_3\bar{R}$



How R-Chart Control Limits Work

Because the standard deviation of the subgroup range σ_R is proportional to $\bar{R}$, the $\pm 3\sigma_R$ limits around $\bar{R}$ factor cleanly into multipliers D_3 and D_4:

1. **Center Line ($\bar{R}$):**

$$\text{CL}_R = \bar{R} = \frac{1}{N}\sum_{i=1}^{N} R_i$$

2. **Control Limits (UCL_R, LCL_R):**

$$\text{UCL}_\mathbf{R} = \mathbf{D_4}\bar{\mathbf{R}}, \qquad \text{LCL}_\mathbf{R} = \mathbf{D_3}\bar{\mathbf{R}}$$

(D_3 and D_4 are statistical constants indexed by subgroup size n.)

Table of D_3 and D_4 Constants & What a Breach Below LCL_R Means

Why Is $D_3 = 0.00$ for $n \leq 6$?

- A batch range $R_i = \max - \min$ can never be negative (≥ 0).
- For small subgroups ($n \leq 6$ vials), $\bar{R} - 3\sigma_R < 0$, so LCL_R is truncated at **0.00**. For $n \geq 7$, $D_3 > 0$.

Discussion: Is $R_i < LCL_R$ Good or Bad in a Clinical Lab?

1. **Genuine Assay Improvement:** Pipette/optical spread truly shrank—validate and lock in SOP!
2. **Frozen Optical Sensor / Dry-Labbing:** Photomultiplier is stuck or a tech is typing identical fake QC numbers!

Subgroup Size (n)	Factor D_3 (LCL_R)	Factor D_4 (UCL_R)
2	0.00	3.27
3	0.00	2.57
4	0.00	2.28
5	**0.00**	**2.11**
6	0.00	2.00
7	0.08	1.92
8	0.14	1.86
9	0.18	1.82
10	0.22	1.78
15	0.35	1.65
20	**0.41**	**1.59**

Step-by-Step R-Chart Limits for Immunoassay Baseline ($N = 6, n = 5$)

Batch	Validated Analyzer Ferritin / Hormone QC Vials ($n = 5$, in ng/mL)					Mean ($\bar{X}$)	Range (R)
1	49.59	53.94	47.11	60.58	43.48	50.94	**17.10**
2	60.41	48.90	29.19	58.87	39.80	47.44	**31.22**
3	65.89	56.96	44.15	60.76	50.39	55.63	**21.74**
4	48.48	51.02	57.36	45.54	63.11	53.10	**17.56**
5	49.10	40.69	42.10	29.26	46.40	41.51	**19.84**
6	42.76	35.11	47.66	46.78	44.93	43.45	**12.55**

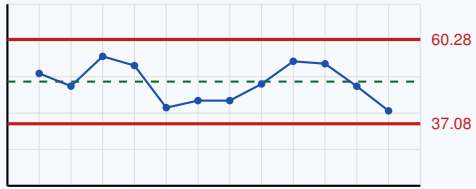
Step-by-Step R-Chart Control Limits ($n = 5$ vials $\implies$ $D_3 = 0.00,\ D_4 = 2.11$)

- **Center Line ($\bar{R}$):** $\bar{R} = \dfrac{17.10 + 31.22 + 21.74 + 17.56 + 19.84 + 12.55}{6} = \mathbf{20.00}$ ng/mL
- **Upper Control Limit ($\mathbf{UCL}_R$):** $\mathrm{UCL}_R = D_4\bar{R} = 2.11 \times 20.00 = \mathbf{42.20}$ ng/mL
- **Lower Control Limit ($\mathbf{LCL}_R$):** $\mathrm{LCL}_R = D_3\bar{R} = 0.00 \times 20.00 = \mathbf{0.00}$ ng/mL

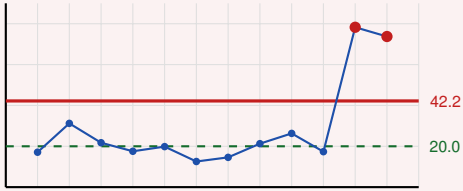
Solving the Edison Batches 101–106 Mystery on the *R*-Chart

Batch	Edison Micro-Cassette Ferritin / Hormone QC Vials ($n = 5$, in ng/mL)					Mean ($\bar{X}$)	Range (R)
101	36.62	50.01	44.39	35.84	50.41	43.45	14.58
102	42.91	34.72	54.82	51.67	56.01	48.03	21.28
103	57.50	49.69	69.49	43.18	51.50	54.27	26.31
104	58.36	53.28	55.59	59.06	41.70	53.60	17.36
105	30.22	16.00	51.87	44.67	94.27	47.40	**78.28 (BREACH!)**
106	2.92	28.45	76.67	66.05	29.04	40.63	**73.75 (BREACH!)**

$\bar{X}$-Chart Alone Looks "In Control"



R-Chart Catches the Edison Dilution Spikes!



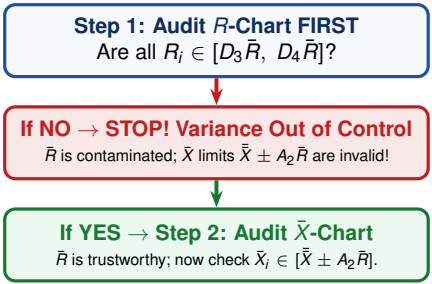
The Golden Rule of SPC: Always Audit the R-Chart BEFORE the $\bar{X}$-Chart!

Why Order of Operations Matters

Look closely at the $\bar{X}$-chart formula:

$$\text{UCL}_{\bar{X}},\ \text{LCL}_{\bar{X}} = \bar{\bar{X}} \pm A_2\bar{\mathbf{R}}$$

- $\bar{R}$ **sits inside the $\bar{X}$-chart control limits!**
- **The "Inflated-Limits Trap":** If an assay has out-of-control variance spikes, $\bar{R}$ becomes artificially huge.
- Multiplying a bloated $\bar{R}$ by A_2 pushes $\text{UCL}_{\bar{X}}$ and $\text{LCL}_{\bar{X}}$ miles apart—making the $\bar{X}$-chart **blind to real shifts in the assay mean**!



Visual Intuition (Before Scenario 3): B12 & Contaminated Limits

WSJ / CMS Forensics: Vitamin B12 Assay

- CMS inspectors discovered that Theranos's Vitamin B12 tests had Coefficients of Variation (CV $= \sigma/\mu$) of **34.3% to 48.5%**, vs. **2.4% to 2.9%** on validated reference analyzers.
- On a 500 pg/mL B12 control ($n = 5$), a validated Siemens analyzer has $\bar{R}_{\text{clean}} = 30.0$ pg/mL.
- Switched to unstable Edison cartridges, daily ranges exploded to $R \in [310,\ 390]$ pg/mL while the mean shifted to 560 pg/mL!

How Pooling Noisy Runs Blinds the $\bar{X}$-Chart

- **Clean Baseline ($\bar{R} = 30.0$):** Half-width is $A_2\bar{R} = 0.58(30) = \pm\mathbf{17.4}$ pg/mL. A mean shift to 560 easily triggers an alarm!
- **Contaminated Pool ($\bar{R}_{\text{pooled}} = 150.0$):** Pooling clean and noisy days bloats $\bar{R}$ by $5\times$ to 150.0 pg/mL.
- Now $A_2\bar{R}_{\text{pooled}} = 0.58(150) = \pm\mathbf{87.0}$ pg/mL—hiding a $+60$ pg/mL mean bias in plain sight!

LAB SESSION 3: Edison B12 Dispersion & Contaminated Limits (Quiz Q13–Q18)

Open Web Simulator Round 3 & Answer Canvas Quiz Q13–Q18

Vitamin B12 QC Audit
($n = 5$ vials/day $\implies$ $A_2 = 0.58,\ D_3 = 0.00,\ D_4 = 2.11$):

- **Q13 (Mode A — Pooled Days 1–8):** What do $\bar{R}_{\text{pooled}}$ (150.0) and $[\text{LCL}_{\bar{X}}, \text{UCL}_{\bar{X}}]$ jump to?
- **Q14 (Contaminated $\bar{X}$-Chart):** How many of the 8 days breach $\text{UCL}_{\bar{X}}^{\text{pooled}} = 598.6$ pg/mL?
- **Q15 (Mode B — Clean Days 1–5 R-Chart):** Compute $\bar{R}_{\text{clean}}$ (30.0) and UCL_R (63.3 pg/mL).
- **Q16 (R-Chart Breaches):** Which Edison days breach $\text{UCL}_R = 63.3$ and what is max R_7?
- **Q17 (Clean $\bar{X}$-Chart Limits):** Under [482.6, 517.4], which 2 Edison days breach $\text{UCL}_{\bar{X}}$?
- **Q18 (Day 8 Nuance):** True/False: Day 8 ($\bar{X}_8 = 488$, $R_8 = 350$) hides inside $\bar{X}$ and is caught only by R?

Web Simulator + 1-Click Gemini AI Prompt (Round 3)

```
Open theranos-sim.html -> Round 3 (Vitamin B12 Dual Audit).
Toggle Mode A (Contaminated Pool Days 1-8) vs Mode B (Clean
Baseline Days 1-5) and toggle the R-Chart Inspector, or click
"Copy Scenario Prompt for Gemini":
Days 1-5 (Siemens):  Xbar=[498,503,496,501,502],
R=[28,32,26,34,30].
Days 6-8 (Edison):  Xbar=[545,560,488], R=[310,390,350].

Compute Mode A and Mode B X-bar and R-chart limits and breaches to

answer Q13--Q18.
```

Under the Hood (Simulator / Sheets / Python)

- **Sheets:** `UCL_R = 2.11*AVERAGE(C2:C6)` | `UCL_X = Xbar + 0.58*Rbar`
- **Python:** `r_c = np.mean(R[:5]); ucl_r = 2.11*r_c; ucl_x = 500 + 0.58*r_c`

FORMULA REMINDER: $n = 5 \implies A_2 = 0.58,\ D_3 = 0,\ D_4 = 2.11$ | $\text{UCL}_R = D_4\bar{R}$ | $\text{UCL}_{\bar{X}},\ \text{LCL}_{\bar{X}} = \bar{\bar{X}} \pm A_2\bar{R}$

Clinical Pathology Forensics: Multi-Channel Pipetting & Finger-Stick Hemolysis

Why Finger-Stick Blood Caused R-Chart Explosions

- **The Medical Mechanism (WSJ Report):** Squeezing a patient's finger to extract drops into a nanotainer ruptures fragile red blood cells (**hemolysis**), spilling intracellular potassium while interstitial tissue fluid dilutes other channels.
- **Multi-Channel Pipette Trap:** When an Edison 6-channel pipette aspirates a hemolyzed/diluted finger-stick sample, micro-clots under-fill Channel 2 while cell lysis spikes Channel 5.
- Replicate average $\bar{X}_i$ looks normal, while within-batch range R_i explodes!

Clinical Lab Root-Cause Diagnostic Guide ($\bar{X}$ vs. R)

Lab Dept.	$\bar{X}$-Chart Breach (Mean Shift μ)	R-Chart Breach (Spread Spike σ)
1. Clinical Chem(Potassium/Ca)	Unrefrigerated reagent decay, calibration offset bias	Finger-stick hemolysis (cell lysis), interstitial dilution
2. Immunoassay(Prolactin/PSA/B12)	Expired antibody lot, incubator temp drift	Robotic micro-pipette air bubbles, saline dilution error
3. Hematology /Coag (PT/INR)	Wrong thromboplastin ISI factor, sensor bias	Micro-clots in nanotainer, erratic plasma centrifugation

When Subgroup Size n Is Large ($n \geq 10$): R-Charts vs. s-Charts

Why Range Loses Efficiency for Large n

- Recall that $R_i = \max(X_{ij}) - \min(X_{ij})$ uses only **2 extreme vials** and ignores the $n-2$ measurements in the middle!
- For small manual QC subgroups ($n = 4$ or 5 vials), R_i is fast and statistically efficient.
- For high-throughput shift panels ($n \geq 10$ or $n = 20$ automated control runs), a single outlier dominates R_i.

Key Features of Larger-n Clinical SPC Charts

- **Non-Zero Lower Range Limit ($n \geq 7$):** For $n \geq 7$, $D_3 > 0$, so $\text{LCL}_R = D_3\bar{R} > 0$ (e.g., at $n = 20$ runs/shift, $D_3 = 0.41,\ D_4 = 1.59$).
- **Modern s-Charts ($n > 10$):** Automated LIMS/Python SPC systems replace R_i with the subgroup standard deviation s_i.
- *Next slide:* Let's audit the real WSJ CMS violation on Theranos's $n = 20$ Coagulation (PT/INR) assay!

In-Class Exercise: Theranos Coagulation Assay QC ($N = 6$ Shifts, $n = 20$)

WSJ / CMS Forensics: Prothrombin INR Index / Clotting Time ($N = 6$ shifts, $n = 20$ control runs/shift)

WSJ Report (p. 5): "The company released 81 patient results for a coagulation test despite frequent quality-control failures." Audit $N = 6$ shifts where each shift runs $\mathbf{n = 20}$ plasma controls (clotting time in minutes / INR index):

Shift (i)	Mean ($\bar{X}_i$)	Range (R_i)
1	3.06	0.42
2	3.15	0.50
3	3.11	0.41
4	3.13	0.46
5	3.06	0.46
6	3.09	0.45

Clinical Audit Questions

1. Which row of the lookup table do you use ($n = 6$ or $n = 20$)?
2. Determine (LCL_R, UCL_R) and ($\text{LCL}_{\bar{X}}$, $\text{UCL}_{\bar{X}}$).
3. Is the coagulation analyzer in statistical control across Shifts 1–6?

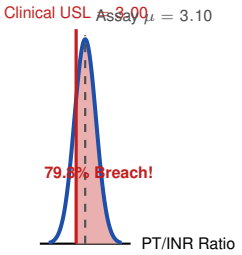
n	A_2	D_3	D_4
6	0.48	0.00	2.00
20	0.18	0.41	1.59

FORMULA REMINDER: $\bar{\bar{X}} = \frac{1}{N}\sum \bar{X}_i$ | $\bar{R} = \frac{1}{N}\sum R_i$ | $\text{LCL}_R,\ \text{UCL}_R = (D_3\bar{R},\ D_4\bar{R})$ | $\text{LCL}_{\bar{X}},\ \text{UCL}_{\bar{X}} = \bar{\bar{X}} \pm A_2\bar{R}$

Bridge Puzzle: Coagulation Assay “In Control” at 3.10 vs. Clinical USL $= 3.00$

The Clinical Quality Paradox (WSJ Coagulation Scandal)

- Our coagulation analyzer is **100% in statistical control**: every shift predictably averages $\mu = 3.10$ with $\sigma = \bar{R}/d_2 = 0.45/3.735 = 0.12$.
- However, the clinical safety upper specification limit for normal plasma control is USL $=$ **3.00** (above which blood-thinner patients on Warfarin risk fatal internal bleeding!).
- **Result:** Over **79.8% of control runs** exceed the USL $= 3.00$ clinical safety ceiling! The analyzer is *in statistical control*, yet *clinically incapable*!



Crucial Clinical Distinction: **Statistical Control** asks: *“Is the analyzer consistent with itself?”* | **Process Capability** asks: *“Does the assay fit inside CLIA patient-safety specifications?”*

Process Capability (C_p and C_{pk}) & Six Sigma

Meeting CLIA Clinical & Regulatory Specification Limits (LSL, USL)

Voice of the Analyzer (LCL, UCL) vs. Voice of the Patient (LSL, USL)

Control Limits [LCL, UCL] = Voice of Analyzer

- **Where they come from:** Computed **from the analyzer's own QC data** ($\bar{\bar{X}} \pm A_2\bar{R} = \mu \pm 3\sigma/\sqrt{n}$).
- **What gets plotted:** Subgroup averages $\bar{X}_i$ and ranges R_i (uses $\sigma_{\bar{X}} = \sigma/\sqrt{n}$).
- **Question answered:** *"Is the analyzer stable and free of assignable shifts over time?"*

Spec Limits [LSL, USL] = Voice of Patient / CLIA

- **Where they come from:** Set **externally** by human physiology and federal FDA/CLIA clinical accuracy laws.
- **What gets evaluated: Individual patient blood tests** X (uses individual assay σ, NEVER $\sigma/\sqrt{n}$!).
- **Question answered:** *"Is each individual patient's blood test clinically safe and accurate?"*

The Theranos "Moving the Goalposts" Scandal (WSJ / CMS Report)

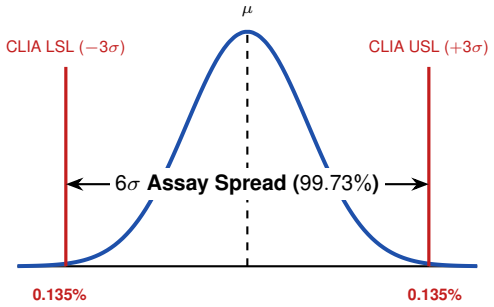
When Edison blood analyzers failed clinical specification limits [LSL, USL], Theranos management **retroactively widened acceptable specification ranges "to match the data"**—illegally overwriting the Voice of the Regulator with the noisy Voice of a broken machine!

Potential Process Capability Ratio (C_p): Assay Spread vs. CLIA Window

Process Capability Ratio (C_p)

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma} = \frac{\text{CLIA Regulatory Tolerance Window}}{\text{Natural } 6\sigma \text{ Assay Spread}}$$

- $\mathbf{C_p < 1.00}$: **Clinically Not Capable** (6σ assay spread is wider than the CLIA tolerance window; $> 0.27\%$ misdiagnoses even if centered).
- $\mathbf{C_p = 1.00}$: **Borderline Capable** ($\text{USL} - \text{LSL} = 6\sigma \implies$ 0.27% or 2,700 DPMO if centered).
- $\mathbf{C_p = 1.33}$: **4-Sigma Clinical Benchmark** ($\text{USL} - \text{LSL} = 8\sigma \implies$ 63 DPMO if centered).
- $\mathbf{C_p = 2.00}$: **Six Sigma Reference Lab** ($\text{USL} - \text{LSL} = 12\sigma$).



Benchmarking C_p Across Five Clinical Assays Audited by CMS (A1–B3)

Clinical Assay Audited	Assay σ	CLIA USL	CLIA LSL	$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}$	Clinically Capable? ($C_p \geq 1$)
Assay A1: Blood Sodium	0.05	100.03	99.97	$\frac{0.06}{6(0.05)} = \mathbf{0.20}$	**No (6σ spread $5\times$ wider than CLIA window!)**
Assay A2: Ionized Calcium	0.01	100.03	99.97	$\frac{0.06}{6(0.01)} = \mathbf{1.00}$	**Borderline ($C_p = 1.00$)**
Assay B1: Blood Gas pH (Siemens)	0.001	5.05	4.95	$\frac{0.10}{6(0.001)} = \mathbf{16.67}$	**Yes (Ultra-Capable Hospital Analyzer)**
Assay B2: Reference Lab pH	0.0005	5.05	4.95	$\frac{0.10}{6(0.0005)} = \mathbf{33.33}$	**Yes (Gold-Standard Reference Lab)**
Assay B3: Edison Point-of-Care pH	0.02	5.05	4.95	$\frac{0.10}{6(0.02)} = \mathbf{0.83}$	**No ($6\sigma = 0.12 > 0.10$ CLIA window)**

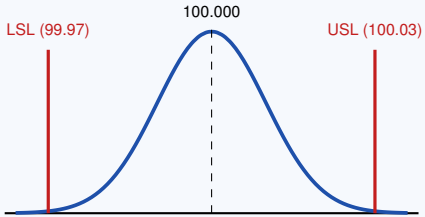
Critical Thinking Question on Assay A2 (Ionized Calcium, $C_p = 1.00$)

Look at the formula $C_p = (\text{USL} - \text{LSL})/(6\sigma)$: **Where is the assay mean μ?** Nowhere! What happens to Assay A2's clinical error rate if its calibration mean shifts off-center from 100.000 to 100.015?

FORMULA REMINDER: $C_p = \frac{\text{USL}-\text{LSL}}{6\sigma}$ | Capable if $C_p \geq 1.00$ (or ≥ 1.33) | Warning: C_p ignores assay centering μ!

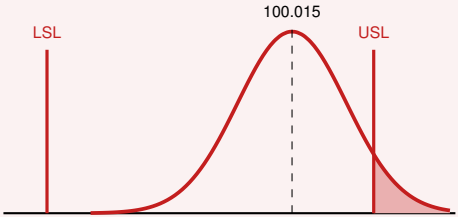
Why C_p Alone Can Lie: Centering Matters! (Assay A2 Shifted $+1.5\sigma$)

$C_p = 1.00$ and Centered ($\mu = 100.000$)



- Below LSL (-3σ): 0.135%
- Above USL ($+3\sigma$): 0.135%
- **Total Clinical Error Rate** $= 0.27\%$ **(**2,700 **DPMO)**

$C_p = 1.00$ but Calibration Shifted $+1.5\sigma$ ($\mu = 100.015$)



- Below LSL (-4.5σ): 0.00034%
- Above USL ($+1.5\sigma$): **6.68%**
- **Total Clinical Error Rate explodes** $25\times$ **to** 6.68%**!**

The assay spread still fits the CLIA tolerance window ($C_p = 1.00$), but off-center calibration pushes 6.68% *of patients past USL!*

Actual Process Capability Index (C_{pk}): Measuring the Nearer CLIA Wall

Process Capability Index (C_{pk})

$$C_{pk} = \min\left\{ \underbrace{\frac{\text{USL} - \mu}{3\sigma}}_{C_{pu}\text{ (Upper CLIA Wall)}}, \underbrace{\frac{\mu - \text{LSL}}{3\sigma}}_{C_{pl}\text{ (Lower CLIA Wall)}} \right\}$$

(Always $\mathbf{C_{pk} \leq C_p}$, with equality iff $\mu = \frac{\text{LSL}+\text{USL}}{2}$)

Assay A2 Centered ($\mu = 100.000$)

Given $\text{USL} = 100.03,\ \text{LSL} = 99.97,\ \sigma = 0.01$:

$$C_{pk} = \min\left\{ \frac{100.03 - 100.00}{3(0.01)}, \frac{100.00 - 99.97}{3(0.01)} \right\}$$
$$= \min\{1.00,\ 1.00\} = \mathbf{1.00}$$

Assay A2 Shifted $+1.5\sigma$ ($\mu = 100.015$)

Assay mean shifts up by 1.5σ to $\mu = 100.015$:

$$C_{pk} = \min\left\{ \frac{100.03 - 100.015}{3(0.01)}, \frac{100.015 - 99.97}{3(0.01)} \right\}$$
$$= \min\{0.50,\ 1.50\} = \mathbf{0.50}\ \text{(Not Capable!)}$$

Fast Z-Score Shortcut: Distance from μ to the nearer CLIA specification limit in standard deviations is simply $\mathbf{Z_{min} = 3C_{pk}}$!

Computing Exact Defect Rates & DPMO (Google Sheets & Python)

Exact Two-Tail Defect Probability Formula

For $X \sim N(\mu, \sigma^2)$ and CLIA specs [LSL, USL]:

1. **Lower Tail Defect ($X <$ LSL):**

$$Z_L = \frac{\text{LSL} - \mu}{\sigma} = -3C_{pl}, \quad P(X < \text{LSL}) = \Phi(Z_L)$$

2. **Upper Tail Defect ($X >$ USL):**

$$Z_U = \frac{\text{USL} - \mu}{\sigma} = +3C_{pu}, \quad P(X > \text{USL}) = \Phi(-Z_U)$$

3. **Total Defect Rate & DPMO:**

$$p_{\text{def}} = \Phi(Z_L) + \Phi(-Z_U), \qquad \text{DPMO} = 10^6 \times p_{\text{def}}$$

Copy-Paste Spreadsheet & Python Recipes

- **Google Sheets (Direct from LSL, USL, μ, σ):**

```
=NORM.DIST(LSL, mu, sig, TRUE) + (1 - NORM.DIST(USL, mu, sig,
TRUE))
```

- **Google Sheets (From C_{pl} and C_{pu}):**

```
=NORM.S.DIST(-3*Cpl, TRUE) + NORM.S.DIST(-3*Cpu, TRUE)
```

- **Python (`scipy.stats.norm`):**

```
from scipy.stats import norm
p_def = norm.cdf(lsl, mu, sig) + norm.sf(usl, mu, sig)
dpmo = 1e6 * p_def
```

LAB SESSION 4: CMS Capability Audit & $100k Portfolio (Quiz Q19–Q24)

Open Web Simulator Round 4 & Answer Canvas Quiz Q19–Q24

Part A: Calcium (LSL = **8.50**, USL = **10.50**) & Part B: $100k Portfolio:

- **Q19 (Preset 2 — Siemens $+1.5\sigma$):** $\mu = 9.75,\ \sigma = 0.167 \implies C_p, C_{pk}$, DPMO?
- **Q20 (Preset 3 — Edison Actual):** $\mu = 9.85,\ \sigma = 0.55 \implies C_p, C_{pk}$, Defect %?
- **Q21 (Preset 4 — Edison Recentered):** $\mu = 9.50,\ \sigma = 0.55 \implies C_{pk}$, Defect %?
- **Q22 (Preset 5 — Fake Specs** [7.50, 12.20]**):** True/False: Fake $C_{pk} = 1.42$ vs. 12.56%?
- **Q23 (Part B — $100k Portfolio):** Which 3 upgrades bring all 3 lines to $C_{pk} \geq 1.33$?
- **Q24 (Part B — Final DPMO):** What does Combined 3-Line DPMO drop to (127.2)?

Web Simulator + 1-Click Gemini AI Prompt (Round 4)

```
Open theranos-sim.html -> Round 4 (CMS Capability & $100k
Portfolio).
Part A: Click Presets 1 to 5 to inspect Cp, Cpk, Defect %, and
DPMO.
Part B ($100,000 Budget across 3 failing lines):
- Line 1 (TSH [90,110], mu=105, sig=2.0):  Recenter mu=100 ($15k)
- Line 2 (Edison Ca [8.5,10.5], mu=9.85, sig=0.55):  Recenter +
Siemens Optics sig=0.25 ($60k)

- Line 3 (Coag PT/INR [10,14], R-chart OOC sig=0.90):  Fix Special

Cause sig=0.50 ($25k).
```

Under the Hood (Simulator / Sheets / Python)

- **Sheets:** `Cp=(10.5-8.5)/(6*sig)` | `Cpk=MIN((10.5-mu)/(3*sig),(mu-8.5)/(3*sig))`
- **Python:** `cp=(usl-lsl)/(6*s); cpk=min((usl-mu)/(3*s),(mu-lsl)/(3*s))`

FORMULA REMINDER: $C_p = \frac{\text{USL}-\text{LSL}}{6\sigma}$ | $C_{pk} = \min\left\{\frac{\text{USL}-\mu}{3\sigma},\ \frac{\mu-\text{LSL}}{3\sigma}\right\}$ | $p_{\text{def}} = \Phi\left(\frac{\text{LSL}-\mu}{\sigma}\right) + \Phi\left(\frac{\mu-\text{USL}}{\sigma}\right)$

Clinical Diagnostic Matrix: Using C_p vs. C_{pk} to Pull the Right Lever

1. High $C_p \geq 1.33$ AND High $C_{pk} \geq 1.33$ (C_{pk}

- **Diagnosis:** Assay spread is narrow **and** centered
- **Executive Lever: Maintain via SPC charts** ($\bar{X}$ ar

2. High $C_p \geq 1.33$ BUT Low $C_{pk} < 1.00$ ($C_{pk} \ll C_p$)

- **Diagnosis:** Assay precision σ is tight enough to fit CLIA specs, but **calibration μ is off-center**!
- **Executive Lever: Fast & Cheap Fix — Recenter μ!** Adjust analyzer calibration offset to $M = \frac{\text{LSL}+\text{USL}}{2}$.

3. Low $C_p < 1.00$ AND Low $C_{pk} \leq C_p < 1.00$

- **Diagnosis: Assay 6σ spread is wider than the CLIA wir**
- **Executive Lever: Hard Biomedical R&D Fix — Shrink σ**

4. Assay Out of Statistical Control (R or $\bar{X}$ Breach)

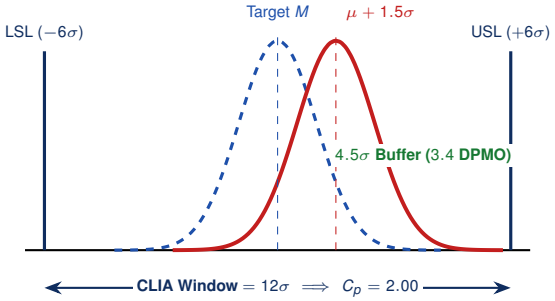
- **Diagnosis:** μ or σ is jumping unpredictably (C_p and C_{pk} formulas are statistically invalid!).
- **Executive Lever: Halt patient testing & eliminate assignable causes first** before computing capability.

Why Six Sigma Means $C_p = 2.00$ and 3.4 DPMO (The 1.5σ Shift)

Solving a Famous Operations & Clinical Lab Puzzle

- Earlier we saw that $\pm 3\sigma$ has 2,700 DPMO, and a perfectly centered $\pm 6\sigma$ assay has only 0.002 DPMO (2 defects per *billion*).
- **Why does every Six Sigma reference lab define Six Sigma as** 3.4 **Defects Per Million (**3.4 **DPMO)?**
- Empirical audits show that even well-controlled analyzers experience reagent/temperature drift of up to $\pm \mathbf{1.5}\sigma$ before $\bar{X}$-charts catch the shift!
- With USL $-$ LSL $= 12\sigma$ ($\mathbf{C_p = 2.00}$) and a 1.5σ shift, the nearer CLIA wall is still $\mathbf{4.5}\sigma$ away ($\mathbf{C_{pk} = 1.50}$):

$$\text{DPMO} = 10^6\big(1 - \Phi(4.5)\big) = \mathbf{3.4}\ \text{DPMO!}$$



“You Make the Call” Executive Poll: Diagnosing Three Clinical Assays

Scenario: CLIA Clinical Tolerance Limits Are [LSL $= 90.0,$ USL $= 110.0$] (Target $M = 100.0$, Window $= 20.0$)

For each hospital/clinical lab assay below, compute C_p and C_{pk} (if valid) and choose the exact clinical intervention:

Assay Alpha (Automated TSH)

- **SPC Status:** Both R and $\bar{X}$ charts are **In Control**.
- **Parameters:** $\mu = \mathbf{105.0}$, $\sigma = \mathbf{2.0}$.
- **Your Call:**
  1. Recenter calibration $\mu \rightarrow 100.0$?
  2. Biomedical R&D to shrink σ?
  3. Hands off (already capable)?

Assay Beta (Edison Testosterone)

- **SPC Status:** Both R and $\bar{X}$ charts are **In Control**.
- **Parameters:** $\mu = \mathbf{100.0}$, $\sigma = \mathbf{5.0}$.
- **Your Call:**
  1. Recenter μ?
  2. Replace/redesign assay to shrink σ?
  3. Hands off (already capable)?

Assay Gamma (Newark Coag PT/INR)

- **SPC Status:** Last 3 shifts breached UCL_R on R-chart!
- **Parameters:** $\hat{\mu} \approx 100.0$.
- **Your Call:**
  1. Compute C_{pk} immediately?
  2. Recenter μ?
  3. Halt patient testing & fix special cause on R-chart first?

FORMULA REMINDER: $C_p = \frac{\text{USL}-\text{LSL}}{6\sigma}$ | $C_{pk} = \min\left\{\frac{\text{USL}-\mu}{3\sigma}, \frac{\mu-\text{LSL}}{3\sigma}\right\}$ | $C_p \geq 1.33 > C_{pk} \implies$ Recenter μ | $C_p < 1 \implies$ Shrink σ

Complete MGT 207 SPC & Process Capability Formula Master Sheet

Part 1: Statistical Process Control (SPC)

- **Subgroup Statistics (N batches of size n vials):**

$$\bar{\bar{X}} = \frac{1}{N}\sum_{i=1}^{N}\bar{X}_i, \quad \bar{R} = \frac{1}{N}\sum_{i=1}^{N}R_i, \quad \hat{\sigma} = \frac{\bar{R}}{d_2}, \quad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

- **Step 1 — Audit R-Chart First (Assay Dispersion):**

$$\text{CL}_R = \bar{R}, \qquad \text{LCL}_R = D_3\bar{R}, \qquad \text{UCL}_R = D_4\bar{R}$$

- **Step 2 — Audit $\bar{X}$-Chart Second (Assay Mean):**

$$\text{CL}_{\bar{X}} = \bar{\bar{X}}, \qquad \text{LCL}_{\bar{X}},\ \text{UCL}_{\bar{X}} = \bar{\bar{X}} \pm A_2\bar{R}$$

- **Golden Rule:** Always look up A_2, D_3, D_4 using **subgroup size** n, never N!

Part 2: Process Capability & Six Sigma

- **Potential Capability Ratio (C_p — Ignores μ):**

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}$$

- **Actual Capability Index (C_{pk} — Penalizes Shift):**

$$C_{pk} = \min\left\{\frac{\text{USL} - \mu}{3\sigma}, \frac{\mu - \text{LSL}}{3\sigma}\right\} \leq C_p$$

- **Exact Defect Probability & DPMO:**

$$p_{\text{def}} = \Phi\left(\frac{\text{LSL} - \mu}{\sigma}\right) + \left[1 - \Phi\left(\frac{\text{USL} - \mu}{\sigma}\right)\right]$$

$$\text{DPMO} = 10^6 \times p_{\text{def}}$$

In-Class Graded Forensics Quiz: Submit All 24 Canvas Questions (Q1–Q24)

Rounds 1 & 2 (Canvas Quiz Q1–Q12)

- **Q1–Q2 (R1 Policy A — $\pm 3\sigma$ Deadband):** Sample SD s, range R, and True/False \$0 cost.
- **Q3–Q4 (R1 Policy B — Full Tampering):** Tampered s, range R, breaches outside [94, 106], and cost.
- **Q5–Q6 (R1 Policy C & Synthesis):** Narrow $\pm 1\sigma$ cost, $\sigma_{\text{tamp}} = \sqrt{2}\sigma$, and Type I α.
- **Q7–Q8 (R2 Phase I Limits & n vs. N Trap):** CL, $\text{UCL}_{\bar{X}}$, $\text{LCL}_{\bar{X}}$ for $n = 4$ vs. false $N = 6$.
- **Q9–Q10 (R2 Breach & Unstopped Penalty):** Exact hour $\bar{X}_t < \text{LCL}_{\bar{X}}$ and 4-hr \$80k CMS penalty.
- **Q11–Q12 (R2 Optimal Andon Cord & Run Rules):** Optimal pull hour (\$3k) and 6-pt trend rule.

Rounds 3 & 4 (Canvas Quiz Q13–Q24)

- **Q13–Q14 (R3 Mode A — Pooled Days 1–8):** Bloated $\bar{R}_{\text{pooled}}$, $[\text{LCL}_{\bar{X}}, \text{UCL}_{\bar{X}}]$, and 0 days flagged.
- **Q15–Q16 (R3 Mode B — Clean R-Chart):** Clean $\bar{R}_{\text{clean}}$, UCL_R, and 3 Edison days breaching R.
- **Q17–Q18 (R3 Clean $\bar{X}$ & Day 8 Nuance):** Clean $\bar{X}$ limits [482.6, 517.4] and why Day 8 needs R.
- **Q19–Q20 (R4 Siemens $+1.5\sigma$ vs. Edison Actual):** C_p, C_{pk}, 3.4 DPMO vs. 12.56% Edison defect rate.
- **Q21–Q22 (R4 Recentered Edison & Fake Specs):** 6.91% recentered defects and fake $C_{pk} = 1.42$.
- **Q23–Q24 (R4 \$100k Portfolio & DPMO):** Optimal 3-line allocation and final 127.2 DPMO.

FORMULA REMINDER: $\sigma_{\text{tamp}} = \sqrt{2}\sigma \mid \text{LCL}_{\bar{X}}, \text{UCL}_{\bar{X}} = \bar{\bar{X}} \pm A_2\bar{R} \mid \text{LCL}_R, \text{UCL}_R = (D_3\bar{R}, D_4\bar{R}) \mid C_p = \frac{\text{USL}-\text{LSL}}{6\sigma} \mid C_{pk} = \min\left\{\frac{\text{USL}-\mu}{3\sigma}, \frac{\mu-\text{LSL}}{3\sigma}\right\}$